FINQUEST INDIA

M1 Deliverable | Problem Statement, Concept, Architecture & Testing Plan

Project overview. FinQuest India is a financial education and planning website for people aged 12 to 60+. Inspired by goal-based financial education platforms, it adapts lessons and tools to a user's life stage while keeping learning, saving and investing concepts clear, accessible and responsible.

1. Problem Statement and Understanding

Many people learn about money informally and may struggle to budget, build savings, understand financial products, compare risk, or plan for major life goals. Financial information can be complex, scattered, or presented in ways that are difficult for beginners. Young learners need safe, age-appropriate financial education; adults may need practical tools for household budgets, saving, investing fundamentals and retirement planning.

Problem statement: Design a trustworthy, accessible, India-focused web platform that helps users of different ages and financial-literacy levels build money-management skills, set realistic goals, explore savings and investment concepts, and practise decisions before making real-world financial choices.

Project understanding and boundaries:

- The platform is educational and planning-focused; it will not take custody of user funds or execute real-money trades in the initial version.
- It must explain that returns are not guaranteed, market-linked investments can lose value, and no financial product is risk-free.
- Recommendations will be educational comparisons based on goals, time horizon, liquidity, costs and risk—not unlicensed or unsupported personalized investment advice.
- Content should reflect Indian financial context, including bank deposits, government schemes, mutual funds, shares, taxation basics, fraud awareness and retirement planning.

2. Idea / Abstract

FinQuest India will turn financial learning into a guided journey. Users select a learning level and goals, then access short lessons, quizzes, calculators and scenario-based activities. A student may practise budgeting with fictional money; a working adult may plan an emergency fund or house deposit; a pre-retiree may estimate retirement expenses. A virtual market simulator will let users explore hypothetical portfolios and observe how risk, diversification, fees and market changes affect outcomes—without investing real money.

Core value proposition: **Learn first. Plan with purpose. Practise safely. Make informed decisions.** The experience will use plain language, responsive design, optional English/Telugu learning content, progress tracking and clear links to authoritative Indian financial-education sources.

3. High-Level Approach / Architecture

LAYER	RESPONSIBILITY
User interface	Responsive home page, age/literacy-level journeys, lessons, dashboard, calculators and accessibility controls.
Learning & planning	Lesson content, quizzes, goals, budgeting logic, savings calculators and simulated market scenarios.
Application / API	Validates inputs, applies calculation rules, retrieves approved content and manages user requests.
Data layer	Stores approved learning content and, if accounts are enabled, minimum necessary profile, progress and saved goals with access controls.
Safety & governance	Age-appropriate safeguards, privacy, consent requirements, secure authentication, content review, disclaimers and audit/error logging.

Initial pipeline: User selects age band and goal → platform identifies suitable learning path → user completes lessons and quizzes → calculator/simulator demonstrates concepts → progress and goal plan are displayed → user is directed to verified official resources when needed.

4. Planned Testing Methodology

Testing will verify that FinQuest is understandable, numerically reliable, secure, accessible and appropriate for its broad age range. Testing will be performed iteratively during development, with issues logged, prioritised and retested after fixes.

TEST TYPE	WHAT WILL BE TESTED	EXAMPLE / SUCCESS CRITERION
Functional testing	Navigation, lesson progression, quizzes, goals, saving plans and dashboard flows.	Links and controls work; progress updates correctly; invalid input produces a helpful message.
Calculation testing	Budget totals, FD/RD illustrations, EMI, inflation, savings targets and retirement estimates.	Compare outputs against independently calculated test cases; handle zero, negative, empty and boundary values safely.
Educational/content testing	Clarity, factual accuracy, age suitability, product descriptions, risks and disclaimers.	Review by a knowledgeable reviewer; users can explain key concepts after lessons; no guaranteed-return claims.
Usability testing	Tasks with users representing younger learners, beginners, working adults and older users.	Users can find a lesson, complete a calculator and save a goal without assistance; feedback informs revisions.
Responsive & accessibility	Mobile, tablet and desktop layouts, keyboard use, readable text, contrast and screen-reader labels.	No clipped content at target screen sizes; core tasks are keyboard-accessible and labels are meaningful.
Security & privacy	Authentication if implemented, input validation, access control, data minimisation and age/guardian safeguards.	Users cannot access another user's saved data; secrets are not exposed; sensitive data is not collected unnecessarily.
Performance & compatibility	Page load, common browsers, broken links and errors on slower connections.	Core pages render reliably on current major browsers and mobile devices; no critical console or server errors.

5. Test Process and Acceptance Criteria

- Prepare test cases for each feature and define expected results before testing.
- Run unit tests for calculation functions and validation rules; run integration tests for connected components and storage where applicable.
- Perform end-to-end checks for the main journeys: learn a concept, complete a quiz, calculate a savings goal and review the result.
- Conduct usability checks with representative age groups, with appropriate guardian consent and safeguards for minors.
- Record defects by severity, fix critical and high-priority issues, and rerun regression tests before release.
- Use only clearly labelled hypothetical examples in simulations; show assumptions, uncertainty and risk disclosures beside financial projections.

Acceptance criteria

The M1 prototype is ready for review when the core user journeys work; financial calculations pass predefined test cases; content and disclosures have been reviewed; the website works on mobile and desktop; no known critical security or privacy defects remain; and feedback from representative users has been documented.

6. Expected Outcome

A scalable prototype for lifelong financial literacy in India—supporting users from age 12 to 60+ through age-appropriate learning, practical budgeting and goal planning, safe investment simulations, and transparent explanations of savings and investment choices.

Scope note: *Real-money transactions, brokerage integration and regulated personalised investment advice are outside the initial scope and would require separate legal, compliance, security and operational review.*